13.5) Directional Derivative

Main topic

1) $D_{\underset{\sim}{u}} f = \underset{\sim}{\nabla} f \cdot \underset{\sim}{u}$; $\underset{\sim}{u}$ is unit $(|\underset{\sim}{u}| = 1)$

2) $\underset{\sim}{\nabla} f = \frac{\partial f}{\partial x}\underset{\sim}{i} + \frac{\partial f}{\partial y} j + \frac{\partial f}{\partial z}\underset{\sim}{k}$ or $\frac{\partial f}{\partial x}\underset{\sim}{i} + \frac{\partial f}{\partial y} j$

(for $n=3$) (for $n=2$)

3) max val of $D_{\underset{\sim}{u}} f = +|\underset{\sim}{\nabla} f|$
achieved in the dir $\underset{\sim}{u} = + \frac{\underset{\sim}{\nabla} f}{|\underset{\sim}{\nabla} f|}$

4) min val of $D_{\underset{\sim}{u}} f = -|\underset{\sim}{\nabla} f|$
achieved in the dir $\underset{\sim}{u} = - \frac{\underset{\sim}{\nabla} f}{|\underset{\sim}{\nabla} f|}$

5) Interpretation of $D_{\underset{\sim}{u}} f [x_0\ y_0]$

- (a) slope of tangent line to the curve on the surface, moving along in the direction $\underset{\sim}{u}$ towards $[x_0\ y_0]$

- (b) Rate of change of f per unit change in the direction $\underset{\sim}{u}$ at point $[x_0\ y_0]$

13.5) Directional Derivatives

20/10/2021

Recall In Sect 13.3.

$$f_x\begin{bmatrix}x_0\\y_0\end{bmatrix} = \frac{\partial f}{\partial x}\begin{bmatrix}x_0\\y_0\end{bmatrix} = \lim_{h\to 0}\frac{f\begin{bmatrix}x_0+h\\y_0\end{bmatrix} - f\begin{bmatrix}x_0\\y_0\end{bmatrix}}{h}$$

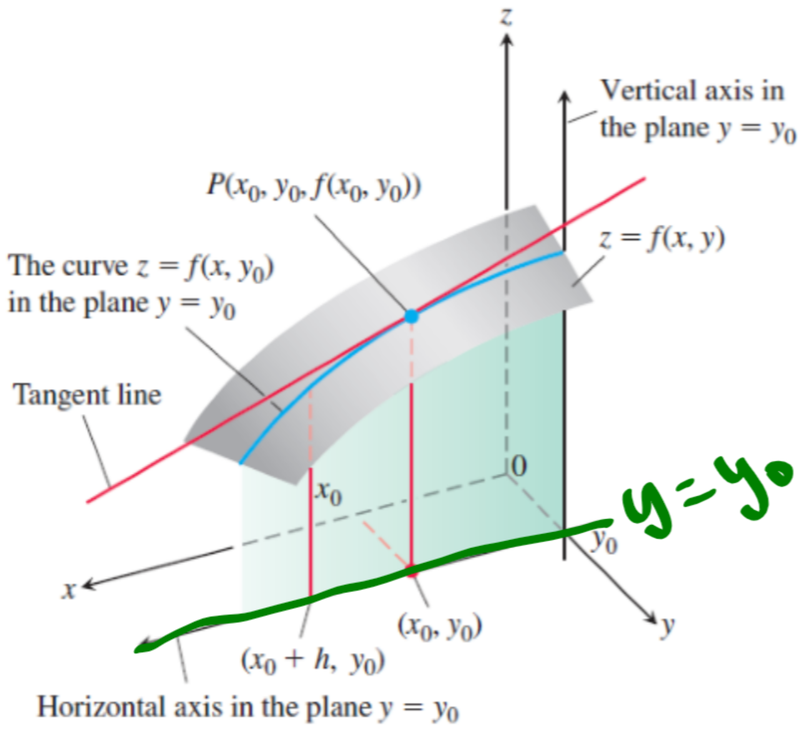


FIGURE 13.16 The intersection of the plane $y = y_0$ with the surface $z = f(x, y)$, viewed from above the first quadrant of the xy-plane.

Fixed $y = y_0$ (const)

(along the line $y = y_0$)

$f_x\begin{bmatrix}x_0\\y_0\end{bmatrix}$ = the slope of the "red" line at $[x_0, y_0)$

$$f_y\begin{bmatrix}x_0\\y_0\end{bmatrix} = \frac{\partial f}{\partial y}\begin{bmatrix}x_0\\y_0\end{bmatrix} = \lim_{h\to 0}\frac{f\begin{bmatrix}x_0\\y_0+h\end{bmatrix} - f\begin{bmatrix}x_0\\y_0\end{bmatrix}}{h}$$

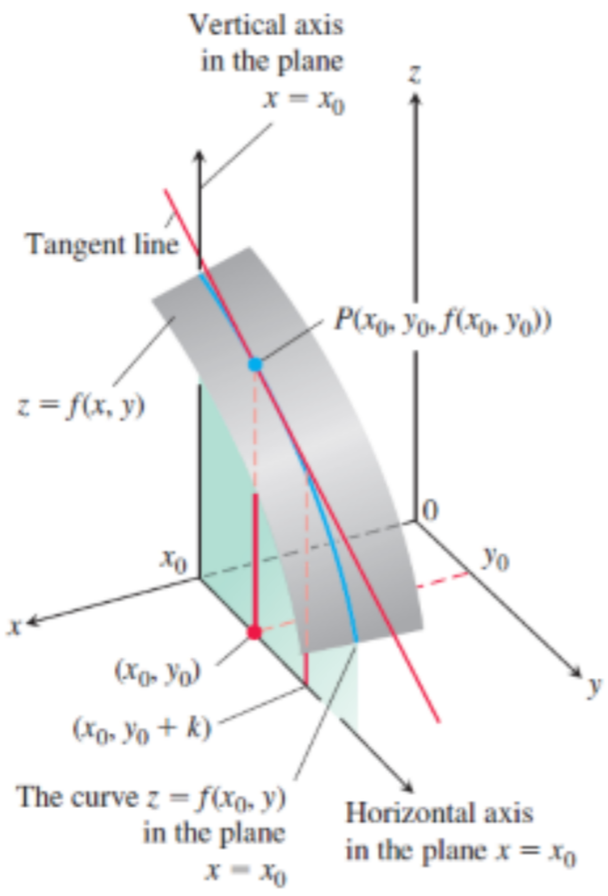


FIGURE 13.17 The intersection of the plane $x = x_0$ with the surface $z = f(x, y)$, viewed from above the first quadrant of the xy-plane.

Fixed $x = x_0$

(Along the line $x = x_0$)

$f_y\begin{bmatrix}x_0\\y_0\end{bmatrix}$ = the slope of the "red line" at pt $[x_0, y_0)$

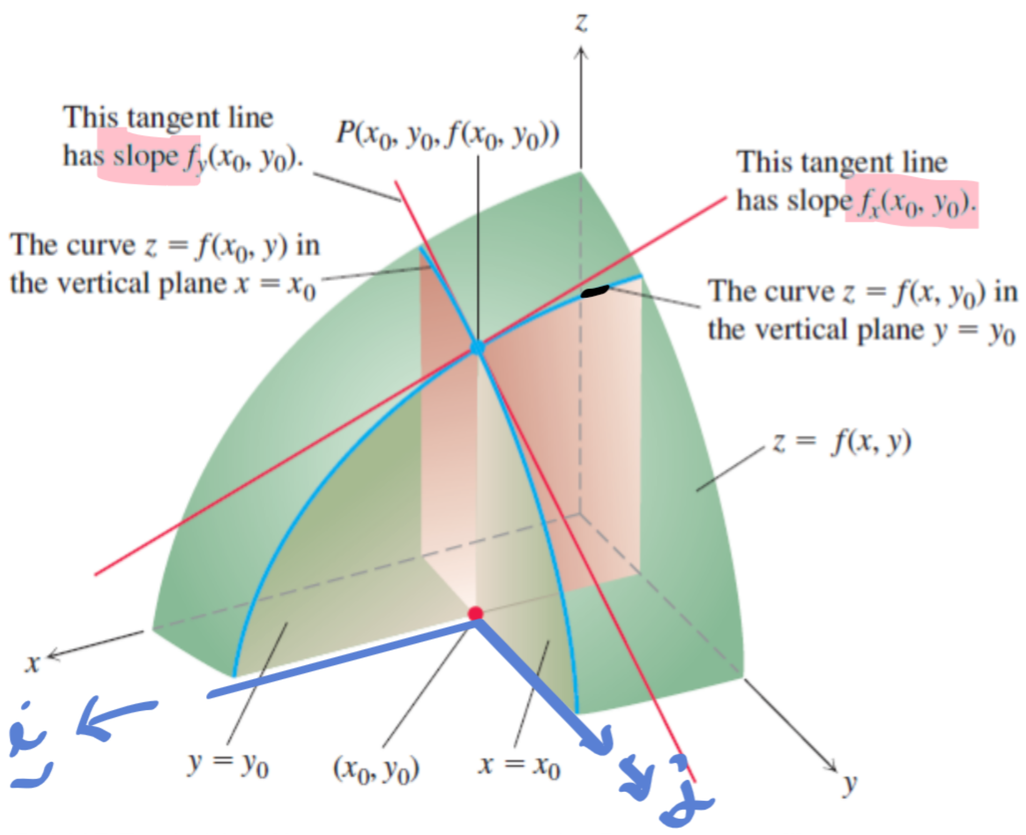
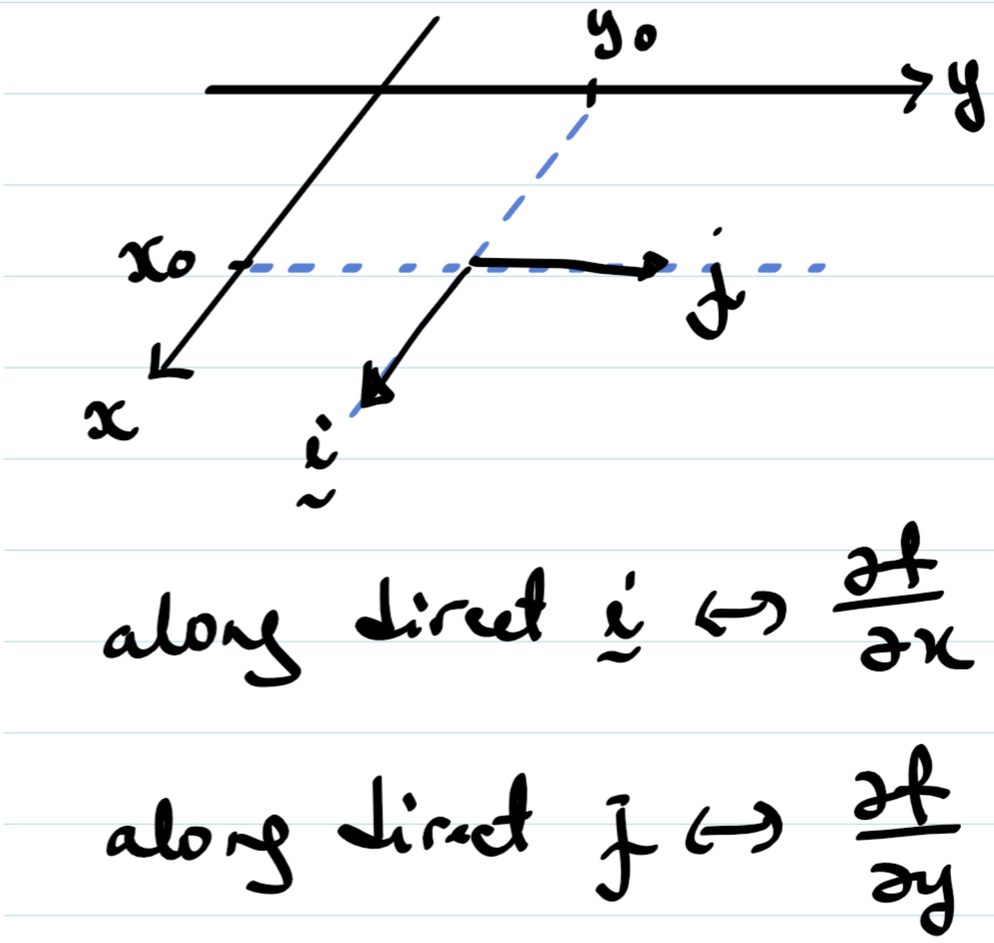


FIGURE 13.18 Figures 13.16 and 13.17 combined. The tangent lines at the point $(x_0, y_0, f(x_0, y_0))$ determine a plane that, in this picture at least, appears to be tangent to the surface.



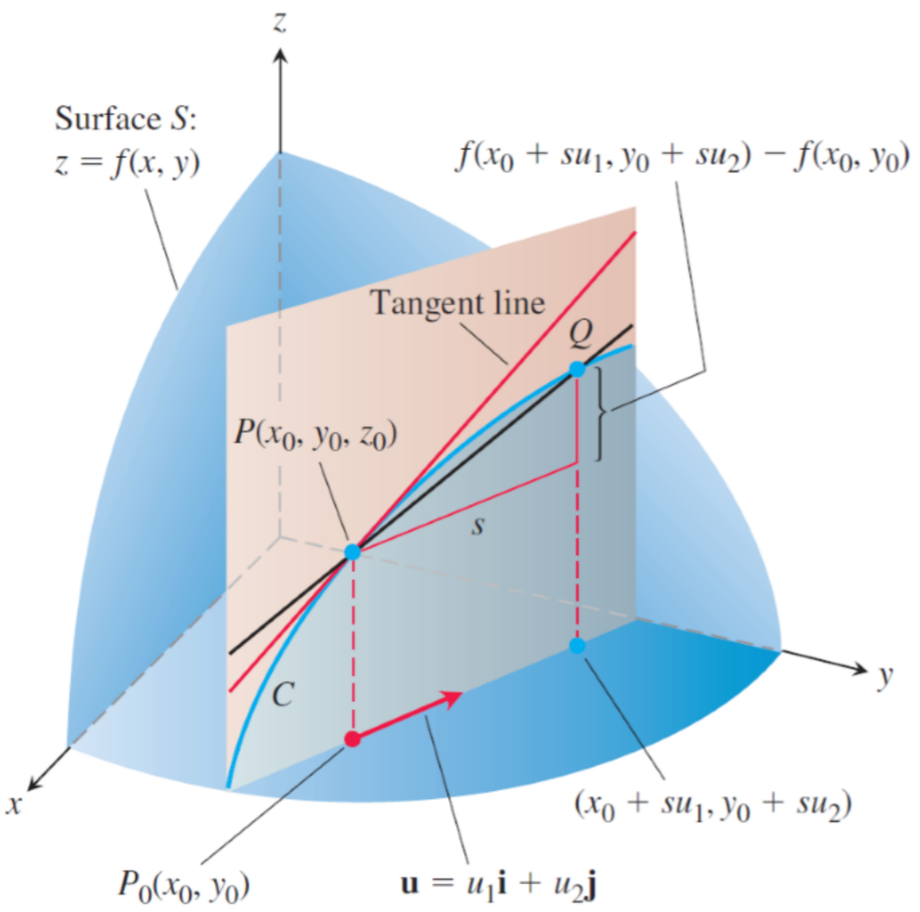
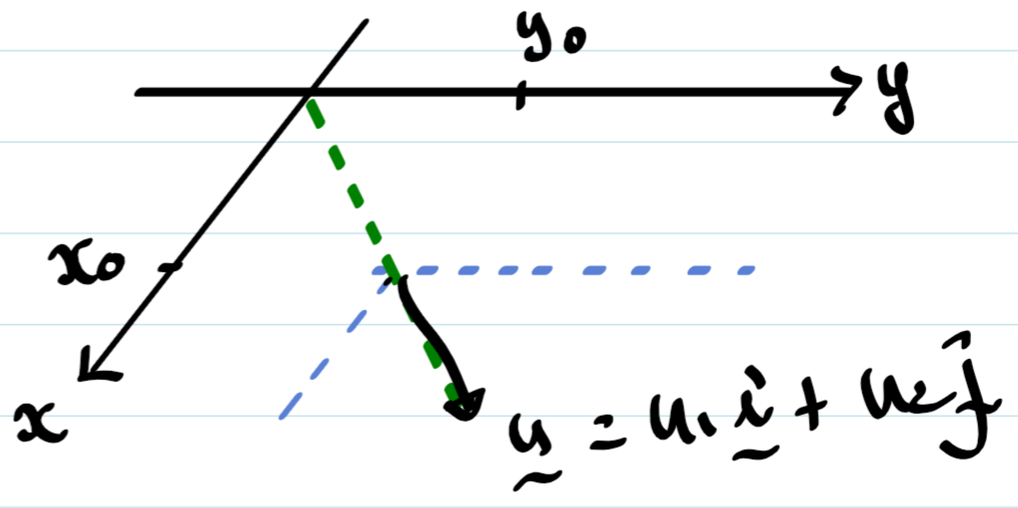
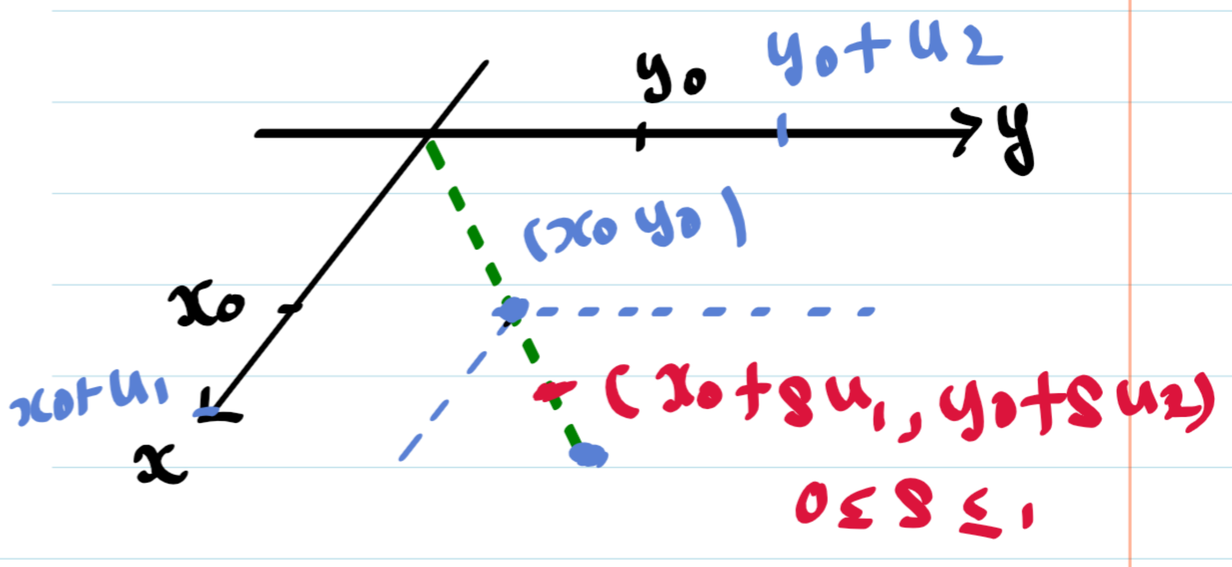


FIGURE 13.28 The slope of the trace curve C at P_0 is $\lim_{Q \to P}$ slope (PQ); this is the directional derivative

$$\left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} = D_{\mathbf{u}} f|_{P_0}.$$





Directional derivative = the slope of the "red line" at the pt $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$ which is produced by moving along the line // $\underset{\sim}{u}$

22/10/2024

a) Dir. Der. - definition using limit

Definition (pp.798: for $n = 2$). [pp.810 in 15th Ed.]

The *directional derivative* of f at point $\underset{\sim}{x_0}$ in the direction of a UNIT vector $\underset{\sim}{u}$ is

$$D_{\underset{\sim}{u}} f(\underset{\sim}{x_0}) = \lim_{s \to 0} \frac{f(\underset{\sim}{x_0} + s\underset{\sim}{u}) - f(\underset{\sim}{x_0})}{s}$$

if this limit exists.

$\underset{\sim}{x_0} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$ $\quad \underset{\sim}{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

$$= \lim_{s \to 0} \frac{f\begin{bmatrix} x_0 + su_1 \\ y_0 + su_2 \end{bmatrix} - f\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}}{s}$$

(i) when $\underset{\sim}{u} = \underset{\sim}{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$D_{\underset{\sim}{u}} f\begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \lim_{s \to 0} \frac{f\begin{bmatrix} x_0 + s \\ y_0 \end{bmatrix} - f\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}}{s} = \frac{\partial f}{\partial x}\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

(ii) when $\underset{\sim}{u} = \underset{\sim}{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\quad u_1 = 0,\ u_2 = 1$

$$D_{\underset{\sim}{u}} f\begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \lim_{s \to 0} \frac{f\begin{bmatrix} x_0 \\ y_0 + s \end{bmatrix} - f\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}}{s} = \frac{\partial f}{\partial y}\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

b) Rmk - Characteristic of $\underset{\sim}{u}$

$\Rightarrow |\underset{\sim}{u}| = 1$

(#) Here $\underset{\sim}{u}$ must be a UNIT vector!!

(#) If $\underset{\sim}{u}$ is NOT unit,

then we have to "normalized" it,

using $\hat{\underset{\sim}{u}} = \dfrac{\underset{\sim}{u}}{|\underset{\sim}{u}|}$ instead of $\underset{\sim}{u}$ in the definition.

eg) for $\underset{\sim}{u} = 5\underset{\sim}{i} - 3\underset{\sim}{j}$, $\quad \hat{\underset{\sim}{u}} = \dfrac{5\underset{\sim}{i} - 3\underset{\sim}{j}}{\sqrt{5^2 + 3^2}} = \dfrac{5}{\sqrt{34}}\underset{\sim}{i} - \dfrac{3}{\sqrt{34}}\underset{\sim}{j}$

b) Dir. Der. – formula using dot product.

Theorem (pp.800: The Directional Derivative is a Dot Product). [pp.812 in 15th Ed.]
Suppose f is a differentiable function. For any unit vector $\underset{\sim}{u} \in \mathbb{R}^n$, $\underset{\sim}{x} \in \mathscr{D}$, we have

$$D_{\underset{\sim}{u}} f(\underset{\sim}{x}) = \nabla f(\underset{\sim}{x}) \cdot \underset{\sim}{u}.$$

eg) $n=2$, $\underset{\sim}{u} = u_1 \underset{\sim}{i} + u_2 \underset{\sim}{j}$

$$D_{\underset{\sim}{u}} f = (f_x \underset{\sim}{i} + f_y \underset{\sim}{j}) \cdot (u_1 \underset{\sim}{i} + u_2 \underset{\sim}{j})$$

$$= u_1 f_x + u_2 f_y .$$

Example. Find the derivative of the function at P_0 in the direction of $\underset{\sim}{u}$.

13.5.18 $h(x,y,z) = \cos xy + e^{yz} + \ln zx, \quad P_0(1,0,1/2), \quad \underset{\sim}{u} = \underset{\sim}{i} + 2\underset{\sim}{j} + 2\underset{\sim}{k}$

(#) Check $|\underset{\sim}{u}| = \sqrt{1+2^2+2^2} = 3 \neq 0$

need to replace with $\hat{\underset{\sim}{u}}$ in the formula !

Soln

$$\underset{\sim}{\nabla} h = h_x \underset{\sim}{i} + h_y \underset{\sim}{j} + h_z \underset{\sim}{k}$$

$$= (-y\sin xy + \frac{1}{x})\underset{\sim}{i} + (-x\sin xy + ze^{yz})\underset{\sim}{j} + (ye^{yz} + \frac{1}{z})\underset{\sim}{k}$$

$$\nabla h(P_0) = \underset{\sim}{i} + \frac{1}{2}\underset{\sim}{j} + 2\underset{\sim}{k}$$

$$\hat{\underset{\sim}{u}} = \frac{\underset{\sim}{u}}{|\underset{\sim}{u}|} = \frac{\underset{\sim}{i} + 2\underset{\sim}{j} + 2\underset{\sim}{j}}{3}$$

$$\therefore D_{\underset{\sim}{u}} h(P_0) = \underset{\sim}{\nabla} h(P_0) \cdot \hat{\underset{\sim}{u}} = \frac{1+1+4}{3} = 2 \neq$$

Theorem 1 (pp.801: Properties of the Directional Derivative). [pp.813 in 15th Ed.]
Given that

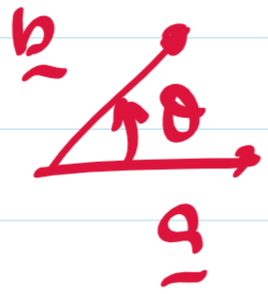
$$D_{\underset{\sim}{u}} f(\underset{\sim}{x}) = \nabla f(\underset{\sim}{x}) \cdot \underset{\sim}{u} = |\nabla f| \cos\theta .$$

(∵ $|\underset{\sim}{u}| = 1$)

(a)
- *The maximum value of the directional derivative* $D_{\underset{\sim}{u}} f(\underset{\sim}{x})$ *is* $|\nabla f(\underset{\sim}{x})|$.
- *It occurs when* $\underset{\sim}{u}$ *has the same direction as the gradient vector* $\nabla f(x)$.

(b)
- *The minimum value of the directional derivative* $D_{\underset{\sim}{u}} f(\underset{\sim}{x})$ *is* $-|\nabla f(\underset{\sim}{x})|$.
- *It occurs when* $\underset{\sim}{u}$ *has the opposite direction as the gradient vector* $\nabla f(x)$.

Rmk Recall in Sim1003



$$\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}| \cos\theta$$

we know $-1 \le \cos\theta \le +1$

max val for $\underset{\sim}{a} \cdot \underset{\sim}{b} \Leftrightarrow \cos\theta = 1 \Leftrightarrow \underset{\sim}{a} \uparrow\uparrow \underset{\sim}{b}$ $(\theta = 0)$

min val for $\underset{\sim}{a} \cdot \underset{\sim}{b} \Leftrightarrow \cos\theta = -1 \Leftrightarrow \underset{\sim}{a} \uparrow\downarrow \underset{\sim}{b}$ $(\theta = \pi)$

[Take $\underset{\sim}{a} = \nabla f$, $\underset{\sim}{b} = \underset{\sim}{u}$]

max " min "

Example. Find the directions in which the functions increase and decrease most rapidly at P_0. Then find the derivatives of the functions in these directions.

13.5.24 $h(x,y,z) = \ln(x^2+y^2-1)+y+6z, \quad P_0(1,1,0)$

Soln $\nabla h = \frac{2x}{x^2+y^2-1}\,\underset{\sim}{i} + \left[\frac{2y}{x^2+y^2-1}+1\right]\underset{\sim}{j} + 6\,\underset{\sim}{k}$

$\nabla h \begin{bmatrix}1\\1\\0\end{bmatrix} = 2\,\underset{\sim}{i} + 3\,\underset{\sim}{j} + 6\,\underset{\sim}{k}$

$|\nabla h(1\,1\,0)| = \sqrt{4+9+36} = \sqrt{49} = 7$

∴ The function increases the most at pt P_0

in the direction of $\underset{\sim}{u} = \frac{2\underset{\sim}{i}+3\underset{\sim}{j}+6\underset{\sim}{k}}{7}$

in the rate $+7$

The function decreases the most at pt P_0

in the direction of $\underset{\sim}{u} = \frac{2\underset{\sim}{i}+3\underset{\sim}{j}+6\underset{\sim}{k}}{-7}$

in the rate -7

(H) Note 1st " 2nd " 3rd object

Given $D_{\underset{\sim}{u}} f = \nabla f \cdot \underset{\sim}{u}$

(i) ∇f and $\underset{\sim}{u}$ are given, then find $D_{\underset{\sim}{u}} f$

(ii) ∇f and $D_{\underset{\sim}{u}} f$ are given, then find $\underset{\sim}{u}$

Example (13.5.30). Let $f(x,y)=\dfrac{x-y}{x+y}$. Find the directions u and the values of $D_{\underset{\sim}{u}}f(-1/2,3/2)$ for which

(e) $D_{\underset{\sim}{u}}f(-1/2,3/2)=1$

Soln $\nabla f = \dfrac{2y}{(x+y)^2}\,\underset{\sim}{i} - \dfrac{2x}{(x+y)^2}\,\underset{\sim}{j}$

$\nabla f(P_0) = 3\,\underset{\sim}{i} + \underset{\sim}{j}$.

write $\underset{\sim}{u} = u_1\,\underset{\sim}{i} + u_2\,\underset{\sim}{j}$

Then $1 = D_{\underset{\sim}{u}}f(P_0) = \nabla f(P_0)\cdot \underset{\sim}{u}$

$1 = 3u_1 + u_2$ ——————— (1)

Since $|\underset{\sim}{u}| = 1$,

$1 = u_1^2 + u_2^2$ ——————— (2)

From (1), (2).

$\underset{\sim}{u} = \underset{\sim}{j}$ or $\underset{\sim}{u} = \dfrac{3}{5}\,\underset{\sim}{i} - \dfrac{4}{5}\,\underset{\sim}{j}$.

Example (13.5.34.). Changing temperature along a circle Is there a direction $\underset{\sim}{u}$ in which the rate of change of the temperature function $T(x,y,z) = 2xy - yz$ (temperature in degrees Celsius, distance in meters) at $P(1,-1,1)$ is $-3°C/m$? Give reasons for your answer.

(#) Find $\underset{\sim}{u}$ so that $D_{\underset{\sim}{u}}T(P) = -3$

Soln Given $T = 2xy - yz$

$$\underset{\sim}{\nabla} T = 2y\,\underset{\sim}{i} + (2x - z)\,\underset{\sim}{j} - y\,\underset{\sim}{k}$$

$$\underset{\sim}{\nabla} T(P) = -2\,\underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$$

$$\therefore \quad |\underset{\sim}{\nabla} T(P)| = \sqrt{6} \cong 2.45$$

It follows that $-\sqrt{6} \leq \text{R.Ch} \leq +\sqrt{6}$

Since Rate of Change must be inside the range $-\sqrt{6}$ to $+\sqrt{6}$, Rate of Change cannot achieve -3 in any direction $\underset{\sim}{u}$.